



Learning cross-task relations for panoptic driving perception

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ABSTRACT

Accurately understanding traffic surroundings is crucial for various autonomous and assisted driving scenarios. The visual perception system must capture the entire scene, including vehicle positions, road conditions, and lane configurations. While existing methods co-train models for these tasks simultaneously, they overlook the topological relationships among roads, lanes, and traffic objects in images. In this paper, we propose leveraging inherent structural relations among these tasks to enhance precise panoptic driving perception. We introduce a cross-task relation mining (CRM) method to achieve this goal. Self-attention mechanisms are used to blend key spatial features within each task, and cross-attention facilitates essential information exchange between tasks, resulting in a more comprehensive scene interpretation. Extensive experiments demonstrate the effectiveness of our approach in complex traffic scenarios.

1. Introduction

Advancements in self-driving and assistance systems emphasize the need to better understand and interpret complex traffic situations, such as recent works [1–3] and so on. Accurately understanding these surroundings is crucial for ensuring the safety, efficiency, and reliability of these systems. Autonomous vehicles and driver assistance technologies rely heavily on the accurate detection and understanding of various elements within the driving scene, including the locations of vehicles, road conditions, and lane configurations.

While significant progress has been made in the field of computer vision [4–6], existing methods primarily focus on training models to simultaneously handle multiple essential visual perception tasks, such as the perception of vehicle locations, the assessment of road drivability, and the recognition of lane lines. Unfortunately, the shortcoming of existing methods is that they focus solely on individual tasks, neglecting the inherent connections between them. Therefore, when faced with difficult examples (e.g., low texture regions, strong exposures, and occlusion), existing state-of-the-art (SOTA) methods may struggle to capture the high-confidence cues required for accurate driving perception.

In a driving scenario, vehicle detection, drivable area segmentation, and lane detection are tightly interconnected in understanding road scenes, as illustrated in Fig. 1. Specifically, (1) Drivable area segmentation is able to assist vehicle detection by excluding non-drivable regions from consideration. For example, sidewalks and pedestrian areas can be eliminated from the search space for vehicles, making the detection process more efficient. (2) Lane detection can refine drivable area

segmentation by providing information about the exact positions of lane boundaries. This information enhances the precision of identifying drivable areas, resulting in a more faithful representation of the road layout. (3) Vehicle detection can assist lane detection by providing additional cues for identifying lanes, especially in challenging scenarios where lane markings might be faded or obscured. In such cases, the presence and positions of vehicles can act as robust references, strengthening the lane detection process. In essence, this collaborative synergy among vehicle detection, drivable area segmentation, and lane detection represents a harmonious partnership that elevates the overall comprehension of road scenes. Therefore, the collaboration between these tasks enhances overall scene understanding and contributes to the goal of safe transportation.

To achieve the above-mentioned idea, we introduce a cross-task relation mining method to elevate the overall understanding of driving scenarios. To begin this process, we establish a backbone and a neck network, serving as the encoder, to extract high-dimensional representations from the input image, where the features are shared for all tasks. Unlike existing methods that straightforwardly decode the shared features into predictions, we construct a multi-scale attention-based interaction module. For each layer of the decoder, self-attentions are strategically employed to adaptively combine task-specific features, thereby leveraging mutual priors that are beneficial to all tasks. Subsequently, the deployment of cross-attention operations allows us to obtain updated task-specific representations. These refined features contribute to more consistent predictions when compared to the initial features. This enhancement is attributed to the harmonious

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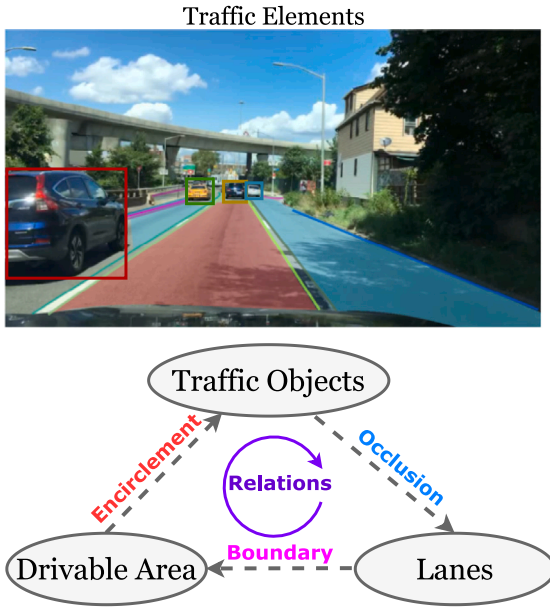


Fig. 1. The structural relations between traffic elements. Predictions for traffic objects, drivable areas, and lanes should not be entirely independent, as these tasks exhibit intricate relations in both 3D space and image space. These cross-task relations create a graph where each task establishes connections with the others.

exchange of complementary information among tasks. Extensive experiments on the BDD100K dataset demonstrate that our method achieves state-of-the-art on all three tasks.

We summarize the contributions of this paper as follows:

1. We present a cross-task relation mining technique that enhances the overall comprehension of driving scenarios by modeling the relations among vehicle detection, drivable area segmentation, and lane detection tasks.
2. Thanks to conventional methods, we propose a multi-scale attention-based interaction module, effectively combining task-specific features through self-attention mechanisms to capture mutual cues across all tasks.
3. Through cross-task complementary information exchange, our method achieves enhanced multi-task consistency and achieves state-of-the-art performance on all tasks.

2. Related work

2.1. Single-task approaches

Traffic Object Detection: Mainstream object detection algorithms can be broadly categorized into two-stage methods and one-stage methods. Two-stage detectors [7–9] predict boxes w.r.t. proposals, whereas single-stage methods make predictions w.r.t. anchors or a grid of possible object centers [10]. Two-stage methods acquire regional proposals, followed by utilizing features within these proposals to accomplish object localization and classification. The evolution of regional proposal generation has seen multiple developmental phases. By contrast, the one-stage methods concurrently perform bounding box regression and object classification. YOLO [11], rather than relying on the RPN network for regional proposals, divides the image into $S \times S$ grids, substantially boosting the detection speed. Incorporating an anchor mechanism, YOLO9000 [12] enhances detection recall. YOLOv4 [13] achieves refinements through the well-designed network structures and loss functions.

Drivable Area Segmentation: Various CNN-based methods have made great progress in semantic segmentation, which also finds practical utility in the drivable area segmentation task, offering outcomes at

the pixel level. As a pioneering work, FCN [14] introduces a fully convolutional network for effective semantic segmentation. PSPNet [15] took a notable stride by introducing the pyramid pooling module, thereby extracting features across multiple scales to enhance its overall performance. For real-time drivable area segmentation, ENet [16] adopted a strategy of reducing the size of feature maps, achieving commendable speed without compromising essential aspects of segmentation. EdgeNet [17] ingeniously fuses edge detection with drivable area segmentation, yielding results that are both more accurate and achieved at a remarkable inference speed.

Lane Detection: Several methods, such as Yang et al. [18], Hou et al. [19], Neven et al. [20], Pizzati et al. [21], Chng et al. [22] and Wu et al. [23], adopt a per-pixel classification approach when tackling lane detection. To address the challenge of classifying lane line points, some previous approaches, like Hou et al. [24], consider each lane line as a distinct category within the segmentation framework. In these methodologies, pixels are classified as either part of the lane or background, resulting in the creation of a binary segmentation mask. Additionally, certain methods go beyond binary segmentation and predict instance segmentation masks, often employing post-clustering strategies. Furthermore, recent advancements, such as those explored in the GraphLane [25], leverage graph structures to model the overall geometry and topology of lane lines, offering enhanced capabilities for handling complex scenarios involving lane fragmentation, merging, and obscured lane markings.

2.2. Multi-task learning approaches

The goal of multi-task learning (MTL) is to enhance representations by sharing information across multiple tasks. Specifically, a convolutional neural network (CNN)-based MTL method can achieve convolutional sharing of the network structure. Mask R-CNN [26] extends Faster R-CNN by adding an object mask prediction branch, effectively combining instance segmentation and object detection tasks, resulting in mutual performance enhancement. Through a shared encoder and three independent decoders, MultiNet [27] concurrently addresses scene classification, object detection, and driving area segmentation tasks. DLT-Net [28] maintains the encoder–decoder structure, fostering context tensor creation between sub-task decoders to facilitate the exchange of pertinent information.

3. Approach

In this section, we first present the basic idea of the proposed method. Secondly, we describe the network architecture in detail. Finally, we elaborate on the overall loss function.

3.1. Overview

As shown in Fig. 2, we compare the pipelines of existing panoptic perception methods and ours. In general, our approach shares a similar network architecture with existing methods, where a shared encoder is employed to extract deep features from images for multiple tasks. Subsequently, we design a specific decoder for each task to generate prediction results. Diverging from conventional approaches, we devise a cross-task relation mining (CRM) module to leverage the inherent structural relations among multiple tasks. We take the decoded multi-scale features as the input of the CRM module to enable the information exchange between task-specific features, building on the intuition that each decoder discovers unique but complementary features due to its separate supervision. The ultimate results of the three tasks will be independently supervised through their respective corresponding ground truth.

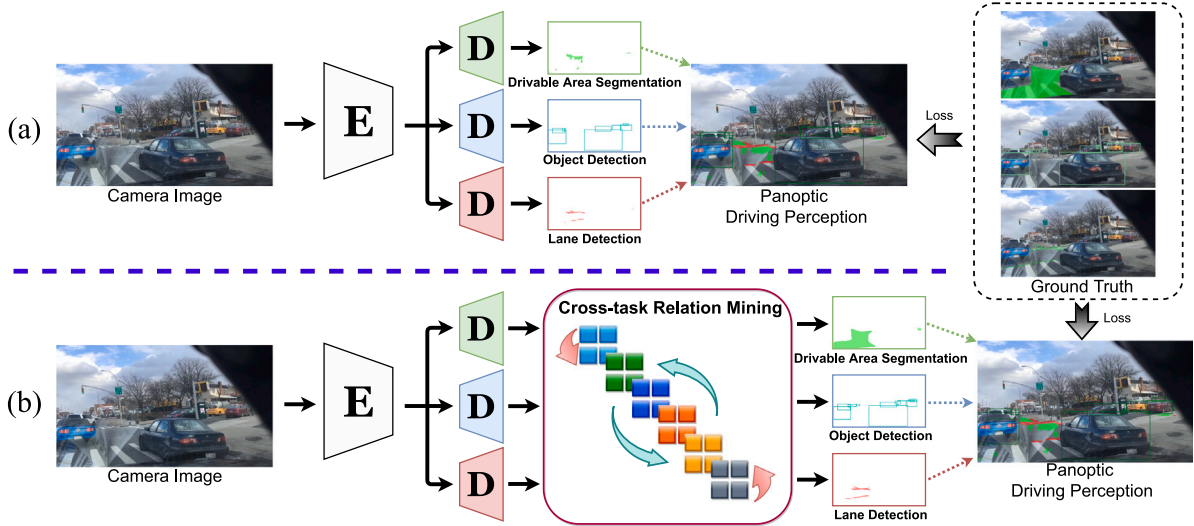


Fig. 2. The pipeline comparison between (a) existing panoptic perception methods and (b) the proposed method. We jointly perform drivable area segmentation, object detection, and lane detection. All tasks share an encoder and have dedicated decoders.

3.2. Encoder

Our encoder contains a shared backbone network and a neck network.

Backbone: The backbone network stands as a pivotal element in the extraction of image input features [29–32], finding widespread application in various domains. Inspired by YOLOv4 [13] and YOLOP [23] in object detection tasks, we adopt CSPDarknet [33] as the backbone for its good trade-off between speed and performance. CSPDarknet has been proven particularly effective in addressing the intricate gradient duplication challenge encountered during optimization [34]. It also boosts the network’s capacity to operate in real-time scenarios, catering to the demands of swift and dynamic processing requirements. Employing a shared backbone network for multi-task learning is a common practice in driving perception methods, contributing to a robust model with several notable benefits. Firstly, it encourages the efficient reuse of features across tasks, reducing computational overhead and enabling faster training and inference. Secondly, it enhances generalization by allowing the model to capture common patterns and dependencies in the data, resulting in improved performance on each task compared to training separate models.

Neck: The neck assumes a crucial function by intricately merging the features derived from the backbone network. Following YOLOP, our neck network is primarily composed of two essential components: the Spatial Pyramid Pooling (SPP) module [35] and the Feature Pyramid Network (FPN) module [36]. The SPP module takes the lead in generating and harmonizing features across a range of scales. In parallel, the FPN module unifies features spanning a spectrum of semantic levels, thereby endowing the resultant features with a rich blend of both scale and semantic depth. This design ensures that the features encapsulate a comprehensive spectrum of information, contributing to the network’s holistic understanding of the input image.

3.3. Decoders

In our network architecture, the three heads serve as dedicated decoders for the respective tasks.

Following YOLOP, we embrace an anchor-based multi-scale detection scheme. Our strategy commences with the utilization of a structure known as the Path Aggregation Network (PAN), which functions as a bottom-up feature pyramid network [37]. By capitalizing on both the top-down semantic features from FPN and the bottom-up positioning features from PAN, we achieve a robust fusion effect for our features.

Subsequently, we employ these multi-scale fusion feature maps for detection within the PAN. Each grid of the multi-scale feature map is associated with three prior anchors boasting distinct aspect ratios. The detection head predicts the positional offset, scaling of height and width, as well as the probability of each category and the confidence of the prediction.

For both the Drivable Area Segment head and the Lane Line Segmentation head, we employ an identical network structure. The segmentation branch is fed with the bottom layer of the FPN, resulting in a $(W/8, H/8, 256)$ feature map. Our segmentation approach is relatively straightforward. By undergoing three successive upsampling stages, we restore the output feature map to the dimensions of $(W, H, 2)$, representing the pixel-wise probability of the input image belonging to the drivable area or lane line category, as well as the background.

In our upsampling layer, we opt for the Nearest Interpolation method to mitigate computational complexity, eschewing the use of deconvolution. This pragmatic choice not only maintains high precision in our segment decoders’ output but also ensures fast inference, striking an optimal balance between performance and efficiency.

3.4. Cross-task relation mining module

Utilizing the multi-scale features extracted from each decoder, we engage in cross-task relation mining for every scale, as shown in Fig. 3. Let $X_k \in \mathbb{R}^{N \times H_k \times W_k \times d_k}$ denote the feature maps acquired from the k th scale. Initially, we enhance each feature with three distinctive learnable positional encodings (PE_N , PE_H , PE_W), representing the task-specific, row-wise, and column-wise indices. Subsequently, we construct Z cross-task self-attention layers to facilitate the exchange of cross-task information. The computational expense for each layer amounts to $\mathcal{O}(H^2 k W^2 k d_k)$. Nonetheless, when dealing with larger feature maps originating from the lower encoder layers, this quadratic term’s cost ($H^2 k W^2 k$) can become prohibitively high. To circumvent the challenges posed by this substantial cost, we introduce a depth-wise separable convolution (DS-Conv) [38] before the attention layers. This DS-Conv effectively condenses the large feature maps into lower-resolution counterparts with the same channel count, denoted as $X'_k \in \mathbb{R}^{N \times h_k \times w_k \times d_k}$. Renowned for its ability to balance efficiency and performance, the DS-Conv offers a more manageable computation load than standard convolutions. To create the pathway across the features of all tasks, we flatten the feature maps into a unified sequence, which includes the elements from all tasks, i.e., $N \times h_k \times w_k$ elements in total.

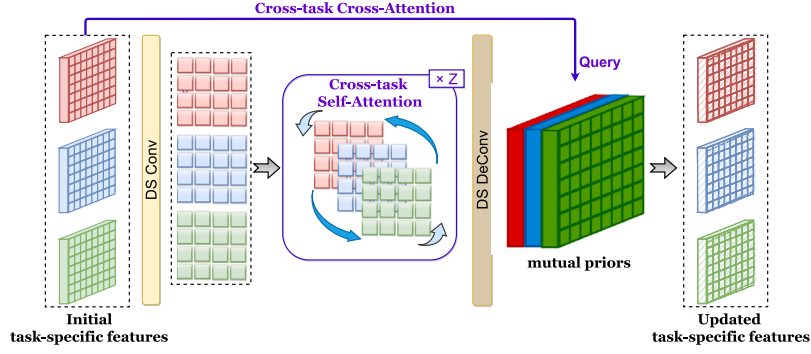


Fig. 3. The illustration of the proposed cross-task relation mining (CRM) module. Our CRM module captures shared knowledge across all tasks, encompassing the entire scene, subsequently generating task-specific features to complement the original ones, thereby leading to more consistent predictions.

Cross-task Self-attention: We introduce three linear layers to derive the query, key, and value vectors from $X'_k + PE_N + PE_H + PE_W$. Subsequently, we partition the features into multiple groups across the channel dimension, enabling multi-head self-attention to enhance feature diversity. This process yields the following representation for the output features:

$$\begin{cases} O_i^{out} = \text{Softmax}(K_i^T Q_i / \sqrt{d_k}) V_i, \\ X^{out} = \text{Concat}(O_1^{out}, \dots, O_M^{out}), \end{cases} \quad (1)$$

where M is the number of feature groups, and K_i, Q_i, V_i denote the i th feature group of the key, query, and value features, respectively. The incorporation of this attention unit brings about a twofold advantage: it encourages each element within X'_k to engage not only with intra-task features, leading to a more profound comprehension of the current frame, but also with inter-task contexts, enabling a comprehensive grasp of the entire scene. To ensure a comprehensive cross-task information exchange, we sequentially stack such units in a cascading manner, iterating Z times. As a counterpart to the flattening process, we reconfigure the outputs of the Z th attention layers into the shape of $N \times h_k \times w_k \times d_k$. Subsequently, we employ a depthwise separable deconvolution to further upsample these features back to the original resolution of the input features.

Cross-task Cross-attention: After acquiring the shared priors of multiple tasks, we proceed to build a cross-attention [39–42] layer that extracts valuable features to refine task-specific characteristics. In contrast to the self-attention layers, the cross-attention takes place at the original scale, with the query vector derived from the initial task-specific features. This approach enables each task to enhance its features through detailed and learned shared priors. Following the completion of cross-task relation mining, the task-specific features are decoded by their respective heads to produce the ultimate outputs.

3.5. Loss function

We basically follow YOLOP to design the loss function. Given the presence of three tasks within our network, our multi-task loss comprises three distinct components. Specifically, the detection loss $\mathcal{L}_{det}$ is formulated as a weighted summation of the classification loss, object loss, and bounding box loss, as depicted in Eq. (2).

$$\mathcal{L}_{det} = \alpha_1 \mathcal{L}_{class} + \alpha_2 \mathcal{L}_{obj} + \alpha_3 \mathcal{L}_{box}, \quad (2)$$

where $\mathcal{L}_{class}$ and $\mathcal{L}_{obj}$ correspond to the focal loss [43], a technique employed to alleviate the loss on well-classified instances, thereby directing the network's attention towards more challenging examples. Here we have set α_1, α_2 , and α_3 to 1.0, 0.8, and 0.8, respectively. Specifically, $\mathcal{L}_{class}$ pertains to the classification penalty, while $\mathcal{L}_{obj}$ addresses prediction confidence. The term $\mathcal{L}_{box}$ pertains to $\mathcal{L}_{CIoU}$ [44], a loss that takes into account factors such as distance, overlap rate, and

similarity in scale and aspect ratio between the predicted box and the ground truth.

Both the drivable area segmentation loss $\mathcal{L}_{da-seg}$ and the lane line segmentation loss $\mathcal{L}_{ll-seg}$ encompass the Cross-Entropy Loss with Logits $\mathcal{L}_{ce}$. This loss mechanism targets the minimization of classification errors between network output pixels and their corresponding targets. It's noteworthy that the $\mathcal{L}_{ll-seg}$ loss incorporates the IoU loss, denoted as $\mathcal{L}_{IoU}$, which is particularly effective for handling the prediction of sparse categories like lane lines. Consequently, $\mathcal{L}_{da}$ and $\mathcal{L}_{ll-seg}$ are defined as shown in Eqs. (3) and (4), respectively.

$$\mathcal{L}_{da-seg} = \mathcal{L}_{ce}, \quad (3)$$

$$\mathcal{L}_{ll-seg} = \mathcal{L}_{ce} + \mathcal{L}_{IoU}. \quad (4)$$

In summary, our overall loss function is represented as the weighted summation of all three components, as illustrated in Eq. (5).

$$\mathcal{L}_{all} = \gamma_1 \mathcal{L}_{det} + \gamma_2 \mathcal{L}_{da-seg} + \gamma_3 \mathcal{L}_{ll-seg}, \quad (5)$$

where the variables $\alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2$, and γ_3 can be adjusted to achieve a balanced integration of all aspects of the overall loss. In the implementation, we have set γ_1, γ_2 , and γ_3 to 1.0, 1.0, 1.0 respectively, and we train our model in an end-to-end manner.

4. Experiments

4.1. Datasets and metrics

Datasets: The BDD100K dataset [45] serves as a valuable resource for advancing multi-task learning within the realm of autonomous driving. Comprising 100,000 frames of images and annotations across 10 distinct tasks, it stands as the most extensive driving video dataset available. This dataset's remarkable diversity in terms of geography, environment, and weather conditions ensures that algorithms trained on it exhibit robustness when applied to new settings. Consequently, we have opted to employ the BDD100K dataset for both the training and evaluation of our network. The BDD100K dataset is partitioned into three segments: a training set containing 70,000 images, a validation set consisting of 10,000 images, and a test set comprising 20,000 images. While the labels for the test set remain undisclosed, our network's evaluation is conducted on the validation set.

Implementation Details: To enhance our detector's understanding of objects within the traffic scene, we follow the practice of YOLOP, using the k-means clustering algorithm to derive prior anchors from the entire set of detection frames within the dataset. During the training process, we employ the Adam optimizer, with initial learning rate l_r , β_1 , and β_2 values set at 0.001, 0.937, and 0.999, respectively. Our training process integrates warm-up and cosine annealing strategies to dynamically adjust the learning rate. Data augmentation is implemented to introduce diversity into the images, thereby fortifying our model

Table 1

Quantitative comparison between the proposed method and state-of-the-art detectors.

Method	Recall (%)	mAP50 (%)	Speed (fps)
MultiNet	81.3	60.2	8.6
DLT-Net	89.4	68.4	9.3
Faster R-CNN	81.2	64.9	8.8
YOLOv5s	86.8	77.2	82
YOLOP	89.2	76.5	41
Our method	91.5	77.7	34

against varying environmental conditions. We employ random rotation, scaling, translation, shearing, and left-right flipping to manipulate images. We conduct image resizing on the BDD100K dataset, transitioning images from a size of $1280 \times 720 \times 3$ to $640 \times 384 \times 3$. The down-sampled feature maps obtained from DS-Conv are 80×48 , and will be upsampled into the original resolution by DS-DeConv. All experiments share identical experimental configurations and evaluation metrics as YOLOP. The experiments are executed using four NVIDIA TITAN XP graphics cards. For drivable area segmentation, both “area/drivable” and “area/alternative” classes in BDD100K dataset are categorized as “Drivable area” without distinction. In our implementation, we have ensured a fair comparison by conducting performance tests of both the proposed approach and baseline networks on the same hardware configuration. We conducted speed tests on an NVIDIA GeForce 3090, and this will be explicitly mentioned in the experimental section. This allows us to provide credible FPS comparison results to support our claims.

4.2. Quantitative results

Traffic Object Detection: Given that MultiNet [27] and DLT-Net [28] are solely focused on vehicle detection, our evaluation concentrates on the vehicle detection outcomes of the several models using the BDD100K dataset, following the practice of YOLOP. As demonstrated in Table 1, we adopt Recall and mAP50 as the key performance indicators for evaluating detection precision. Our model demonstrates superior performance over Faster R-CNN, MultiNet, DLT-Net, YOLOv5s, and YOLOP in terms of detection accuracy. Notably, our model achieves real-time inference capabilities.

Although our approach exhibits a slightly reduced processing speed compared to YOLOP, we have effectively boosted object detection performance through the synergistic exchange of information among multiple tasks. Meanwhile, YOLOv5s attains the fastest processing speed among the existing models due to its architecture not incorporating the lane line segment head and drivable area segment head.

Drivable Area Segmentation: In the experiments, the drivable area segmentation task was characterized as a binary segmentation objective. Specifically, our model’s primary task entailed discerning solely between the drivable area and the background within the given image. To gauge the effectiveness of various models in performing this segmentation, the metric of mean Intersection over Union (mIoU) was employed. The corresponding outcomes are illustrated in Table 2. Evidently, our model showcases a marked superiority compared to MultiNet, DLT-Net, PSPNet, and YOLOP, surpassing them by margins of 19.9%, 20.2%, 1.9%, and 1.9%, respectively. This signifies our model’s exceptional competence in accurately identifying the drivable areas within images.

Lane Detection: The experimental parameters are designed in accordance with the protocols established in [19,23] to facilitate seamless comparisons. In the initial steps, center lines are computed leveraging the dual-line annotations. Subsequently, the lane lines for training are delineated with a width set at 8 pixels, while the lane line width for the test set remains at 2 pixels. Evaluation of model performance is conducted through metrics of pixel accuracy and Intersection over Union (IoU) pertaining to the lanes. As delineated in Table 3, our model unequivocally outperforms the other four models, demonstrating a substantial enhancement in lane line detection capabilities.

Table 2

Quantitative comparison between the proposed method and state-of-the-art drivable area segmentation methods.

Network	mIoU (%)	Speed (fps)
MultiNet	71.6	8.6
DLT-Net	71.3	9.3
PSPNet	89.6	11.1
YOLOP	91.5	41
Our method	93.4	34

Table 3

Quantitative comparison between the proposed method and state-of-the-art lane detection methods.

Network	Accuracy (%)	IoU (%)	Speed (fps)
ENet	34.12	14.64	100
SCNN	35.79	15.84	19.8
ENet-SAD	36.56	16.02	50.6
YOLOP	70.50	26.20	41
Our method	72.07	28.01	34



Fig. 4. The qualitative results of the proposed method and the state-of-the-art method YOLOP, where the predictions of all tasks are plotted together.

Table 4

Ablation study of the proposed cross-task relation mining module. “Det” corresponds to object detection, “DA” relates to drivable area segmentation, and “LD” stands for lane detection.

Baseline	SA	CA	Det	DA	LD
✓			76.5	91.5	70.5
✓	✓		76.9	92.5	70.7
✓		✓	76.3	92.4	71.0
✓	✓	✓	77.7	93.4	72.1

4.3. Qualitative results

Fig. 4 shows the visualization of the predicted results of our method and YOLOP. From the result comparison, the predictions of our method are more accurate than YOLOP. They also align better with the prior knowledge of driving scenarios. This can be attributed to our explicit modeling of structural relationships among multiple tasks.

4.4. Ablation studies

Effect of Cross-task relation mining module: We conduct ablation experiments on the BDD100K [45] dataset to verify the effectiveness

Table 5
Ablation study of different training strategies.

Training method	mAP50	mIoU	Acc.	Speed (ms)
Det (only)	76.9	–	–	15.7
DA (only)	–	92.0	–	14.8
LD (only)	–	–	79.6	14.8
YOLOP (multi)	76.5	91.5	70.5	24.4
Our method (multi)	77.7	93.4	72.1	28.8

of each component. Table 4 illustrates an ablation study conducted on the proposed cross-task relation mining module. Each row corresponds to a specific model configuration, while the columns represent distinct modules.

The “SA” column signifies the presence of a self-attention module. The first row demonstrates performance without self-attention, while the second row exhibits the performance gain with the self-attention module. This suggests that the self-attention module contributes to capturing long-range dependencies in the input data. The “CA” column indicates the inclusion of a cross-attention module. This implies that integrating the cross-attention module positively influences performance, due to its capability to capture relationships between tasks.

Both the SA and CA modules enhance performance across tasks, suggesting their importance in modeling inter-task relationships. Incorporating both SA and CA modules together leads to a larger performance improvement compared to using them individually. This indicates that these modules complement each other, capturing relationships at different levels.

Effect of Multi-task training strategy: We conduct an ablation study to compare various training strategies, investigating their impact on the performance of panoptic driving perception. As shown in Table 5, the strategies include single-task training strategies focused on object detection (Det), drivable area segmentation (DA), and lane detection (LD), as well as multi-task methods like YOLOP and the proposed method. Notable findings indicate that our approach achieves the highest object detection and drivable area segmentation scores, demonstrating its superior effectiveness in simultaneously handling multiple relative tasks. Additionally, our multi-task strategy outperforms YOLOP across multiple metrics, although with a slightly slower inference speed. This proves that the proposed approach strikes a favorable balance between accuracy and speed, making it promising for multi-task panoptic driving perception.

5. Conclusion

In this paper, we have introduced an end-to-end approach to enhance the precision of driving scene perception. By leveraging the inherent relationships among tasks like vehicle detection, drivable area segmentation, and lane detection, our Cross-Task Relation Mining (CRM) method enables a more comprehensive understanding of complex traffic scenarios. Through self-attention mechanisms and cross-attention, we improve the integration of spatial features and critical information exchange between tasks. Our approach demonstrates its effectiveness in achieving a refined panoptic driving perception, marking a promising contribution to intelligent driving systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Y. Zhang, Y. Su, J. Yang, J. Ponce, H. Kong, When Dijkstra meets vanishing point: a stereo vision approach for road detection, *IEEE Trans. Image Process.* (2018) 2176–2188.
- [2] Z. Zheng, X. Li, Q. Xu, X. Song, Deep inference networks for reliable vehicle lateral position estimation in congested urban environments, *IEEE Trans. Image Process.* (2021) 8368–8383.
- [3] Y. Wu, L. Zhao, J. Lu, H. Yan, Dense hybrid proposal modulation for lane detection, 2023, *CoRR abs/2304.14874*.
- [4] Z. Wang, J. Lu, Z. Wu, J. Zhou, Learning efficient binarized object detectors with information compression, *IEEE Trans. Pattern Anal. Mach. Intell.* (2021) 3082–3095.
- [5] X. Xu, Z. Wang, J. Zhou, J. Lu, Binarizing sparse convolutional networks for efficient point cloud analysis, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 5313–5322.
- [6] Z. Wu, C. Song, H. Yan, TC-net: Transformer-convolutional networks for road segmentation, in: *2022 IEEE International Conference on Multimedia and Expo (ICME)*, 2022, pp. 01–06.
- [7] R. Girshick, J. Donahue, T. Darrell, J. Malik, Rich feature hierarchies for accurate object detection and semantic segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2014, pp. 580–587.
- [8] J. Dai, Y. Li, K. He, J. Sun, R-FCN: object detection via region-based fully convolutional networks, in: *Advances in Neural Information Processing Systems*, 2016, pp. 379–387.
- [9] C. Han, Q. Zhao, S. Zhang, Y. Chen, Z. Zhang, J. Yuan, YOLOv2: Better, faster, stronger for panoptic driving perception, 2022, *CoRR abs/2208.11434*.
- [10] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, SSD: Single shot multibox detector, in: *European Conference on Computer Vision*, Springer, 2016, pp. 21–37.
- [11] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: Unified, real-time object detection, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788.
- [12] J. Redmon, A. Farhadi, YOLO9000: better, faster, stronger, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 7263–7271.
- [13] A. Bochkovskiy, C. Wang, H.M. Liao, YOLOv4: Optimal speed and accuracy of object detection, 2020, *CoRR abs/2004.10934*.
- [14] J. Long, E. Shelhamer, T. Darrell, Fully convolutional networks for semantic segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 3431–3440.
- [15] H. Zhao, J. Shi, X. Qi, X. Wang, J. Jia, Pyramid scene parsing network, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2881–2890.
- [16] A. Paszke, A. Chaurasia, S. Kim, E. Culurciello, ENet: A deep neural network architecture for real-time semantic segmentation, 2016, *CoRR abs/1606.02147*.
- [17] H.-Y. Han, Y.-C. Chen, P.-Y. Hsiao, L.-C. Fu, Using channel-wise attention for deep CNN based real-time semantic segmentation with class-aware edge information, *IEEE Trans. Intell. Transp. Syst.* (2020) 1041–1051.
- [18] J. Yang, L. Zhang, H. Lu, Lane detection with versatile atrousformer and local semantic guidance, 2022, *CoRR abs/2203.04067*.
- [19] Y. Hou, Z. Ma, C. Liu, C.C. Loy, Learning lightweight lane detection CNNs by self attention distillation, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1013–1021.
- [20] D. Neven, B. De Brabandere, S. Georgoulis, M. Proesmans, L. Van Gool, Towards end-to-end lane detection: an instance segmentation approach, in: *2018 IEEE Intelligent Vehicles Symposium (IV)*, 2018, pp. 286–291.
- [21] F. Pizzati, M. Allodi, A. Barrera, F. García, Lane detection and classification using cascaded CNNs, in: *EUROCAST*, 2019, pp. 95–103.
- [22] Z.M. Chng, J.M.H. Lew, J.A. Lee, RONELDV2: A faster, improved lane tracking method, 2022, *CoRR abs/2202.13137*.
- [23] D. Wu, M. Liao, W. Zhang, X. Wang, YOLOP: you only look once for panoptic driving perception, 2021, *CoRR abs/2108.11250*.
- [24] Y. Hou, Z. Ma, C. Liu, T.-W. Hui, C.C. Loy, Inter-region affinity distillation for road marking segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 12483–12492.
- [25] P. Lu, S. Xu, H. Peng, Graph-embedded lane detection, *IEEE Trans. Image Process.* (2021) 2977–2988.

- [26] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2961–2969.
- [27] M. Teichmann, M. Weber, J.M. Zöllner, R. Cipolla, R. Urtasun, MultiNet: Real-time joint semantic reasoning for autonomous driving, in: Intelligent Vehicles Symposium, IEEE, 2018, pp. 1013–1020.
- [28] Y. Qian, J.M. Dolan, M. Yang, DLT-Net: Joint detection of drivable areas, lane lines, and traffic objects, *IEEE Trans. Intell. Transp. Syst.* (2019) 4670–4679.
- [29] W. Zheng, J. Lu, J. Zhou, Hardness-aware deep metric learning, *IEEE Trans. Pattern Anal. Mach. Intell.* (2021) 3214–3228.
- [30] W. Zheng, J. Lu, J. Zhou, Deep metric learning with adaptively composite dynamic constraints, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [31] C. Wang, W. Zheng, Z. Zhu, J. Zhou, J. Lu, Introspective deep metric learning, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [32] C. Wang, W. Zheng, Z. Zhu, J. Zhou, J. Lu, OPERA: omni-supervised representation learning with hierarchical supervisions, in: ICCV, 2023, pp. 5559–5570.
- [33] C. Wang, A. Bochkovskiy, H.M. Liao, Scaled-YOLOv4: Scaling cross stage partial network, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2021, pp. 13029–13038.
- [34] C.-Y. Wang, H.-Y.M. Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, I.-H. Yeh, CSPNet: A new backbone that can enhance learning capability of CNN, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2020, pp. 390–391.
- [35] K. He, X. Zhang, S. Ren, J. Sun, Spatial pyramid pooling in deep convolutional networks for visual recognition, *IEEE Trans. Pattern Anal. Mach. Intell.* (2015) 1904–1916.
- [36] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 936–944.
- [37] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 8759–8768.
- [38] A.G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, MobileNets: Efficient convolutional neural networks for mobile vision applications, 2017, CoRR abs/1704.04861.
- [39] B. Yu, W. Li, X. Li, J. Lu, J. Zhou, Frequency-aware spatiotemporal transformers for video inpainting detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8188–8197.
- [40] L. Zhao, W. Zheng, Y. Duan, J. Zhou, J. Lu, SPTR: Structure-preserving transformer for unsupervised indoor depth completion, *IEEE Trans. Circuits Syst. Video Technol.* (2023).
- [41] L. Zhao, J. Lu, J. Zhou, Similarity-aware fusion network for 3D semantic segmentation, in: 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021, pp. 1585–1592.
- [42] Z. Song, L. Zhao, J. Zhou, Learning hybrid semantic affinity for point cloud segmentation, *IEEE Trans. Circuits Syst. Video Technol.* (2021) 4599–4612.
- [43] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2980–2988.
- [44] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, D. Ren, Distance-IOU loss: Faster and better learning for bounding box regression, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2020, pp. 12993–13000.
- [45] F. Yu, W. Xian, Y. Chen, F. Liu, M. Liao, V. Madhavan, T. Darrell, BDD100K: A diverse driving video database with scalable annotation tooling, 2018, CoRR abs/1805.04687.